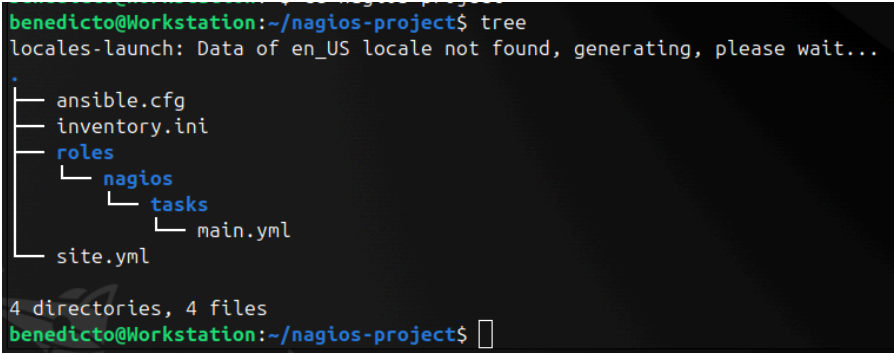
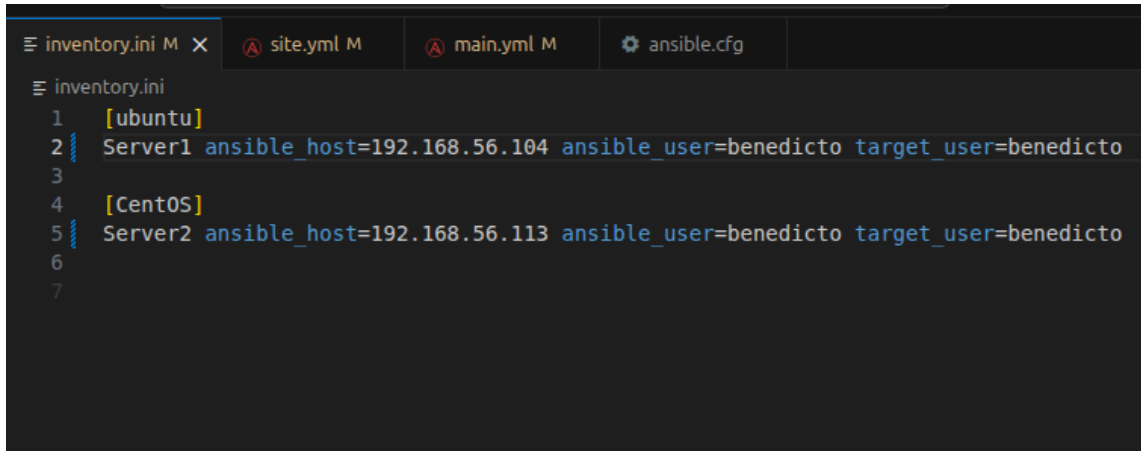


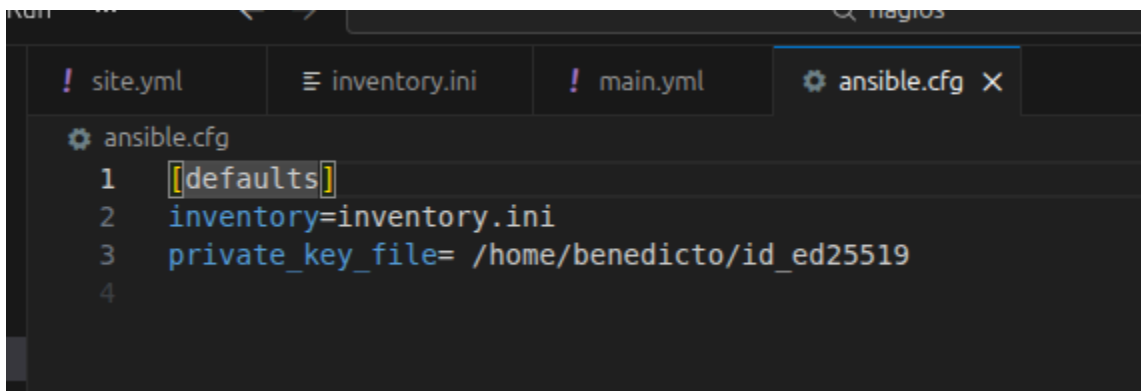
Name: Benedicto, Ruel Jr A.	Date Performed: March 5, 2026
Course/Section: CPE212-CPE31S1	Date Submitted: March 5, 2026
Instructor: Engr. Robin Valenzuela	Semester and SY: 2nd Semester 2025 - 2026
Activity 8: Install, Configure, and Manage Availability Monitoring tools	
1. Objectives	
Create and design a workflow that installs, configure and manage enterprise monitoring tools using Ansible as an Infrastructure as Code (IaC) tool.	
2. Discussion	
Availability monitoring is a type of monitoring tool that we use if the certain workload is up or reachable on our end. Site downtime can lead to loss of revenue, reputational damage and severe distress. Availability monitoring prevents adverse situations by checking the uptime of infrastructure components such as servers and apps and notifying the webmaster of problems before they impact on business.	
3. Tasks	
1. Create a playbook that installs Nagios in both Ubuntu and CentOS. Apply the concept of creating roles. 2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.) 3. Show an output of the installed Nagios for both Ubuntu and CentOS. 4. Make sure to create a new repository in GitHub for this activity.	
4. Output (screenshots and explanations)	
First thing to do is to create a directory only for this and I named it nagios. (It changed to nagios-project because I copied the code from github and I used the TipPC.) After that I created a directory inside the nagios directory and named it roles and under roles I added nagios/tasks/main.yml. I added site.yml where I will put the roles.	
 <pre> benedicto@Workstation:~/nagios-project\$ tree locales-launch: Data of en_US locale not found, generating, please wait... . ├── ansible.cfg ├── inventory.ini ├── roles │ ├── nagios │ │ └── tasks │ │ └── main.yml └── site.yml 4 directories, 4 files benedicto@Workstation:~/nagios-project\$ </pre>	

Second, I added the inventory.ini and the ansible.cfg, probably the most important all in the same nagios directory.



The screenshot shows a code editor with four tabs: inventory.ini, site.yml, main.yml, and ansible.cfg. The inventory.ini file is open and contains the following content:

```
1 [ubuntu]
2 Server1 ansible_host=192.168.56.104 ansible_user=benedicto target_user=benedicto
3
4 [CentOS]
5 Server2 ansible_host=192.168.56.113 ansible_user=benedicto target_user=benedicto
6
7
```



The screenshot shows a code editor with four tabs: site.yml, inventory.ini, main.yml, and ansible.cfg. The ansible.cfg file is open and contains the following content:

```
1 [[defaults]]
2 inventory=inventory.ini
3 private_key_file= /home/benedicto/id_ed25519
4
```

After doing lots of activities I encountered mistakes and possible mistakes in doing the inventory.ini and the ansible.cfg that's why it's crucial to memorize it all perfectly.

The third step is to create the code in the main.yml under roles/nagios/tasks like in the figure below.

```
inventory.ini M  site.yml M  main.yml M X  ⚙️
es > nagios > tasks > main.yml > {} 4 > {} service > abc enable
Ansible Tasks Schema - Ansible tasks file (ansible.json)
1  ---
2
3  - name: Install EPEL repository (CentOS)
4    dnf:
5      name: epel-release
6      state: present
7      when: ansible_os_family == "RedHat"
8
9  - name: Install Nagios on Ubuntu
10   apt:
11     name: nagios4
12     state: present
13     update_cache: yes
14     when: ansible_os_family == "Debian"
15
16  - name: Install Nagios on CentOS
17     dnf:
18       name: nagios
19       state: present
20       when: ansible_os_family == "RedHat"
21
22  - name: Enable and Start Nagios on Ubuntu
23     service:
24       name: nagios4
25       state: started
26       enabled: yes
27       when: ansible_os_family == "Debian"
28
29  - name: Enable and Start Nagios on CentOS
30     service:
31       name: nagios
32       state: started
33       enabled: yes
34       when: ansible_os_family == "RedHat"
```

With the help of AI I was able to do the codes and it's lots of trial and error.

```

inventory.ini M
site.yml M
main.yml M
ansible

site.yml > ...
Ansible Playbook - Ansible playbook files (ansible.json)
1 ---
2 - name: Install and configure nagios
3   hosts: all
4   become: true
5
6   roles:
7     - nagios
8
9

```

After that, I create a role that will call the nagios and run the modules using ansible-playbook site.yml -K command.

And lastly, I opened Ubuntu and centOS to verify installation and services.

Ubuntu

[illegible]

CentOS

```
benedicto@vbox:~ — systemctl status nagios

● nagios.service - Nagios Core 4.4.14
   Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: d>
   Active: active (running) since Thu 2026-03-05 18:08:38 PST; 40s ago
     Docs: https://www.nagios.org/documentation
   Process: 6364 ExecStartPre=/usr/sbin/nagios -v /etc/nagios/nagios.cfg (code>
   Process: 6365 ExecStart=/usr/sbin/nagios -d /etc/nagios/nagios.cfg (code=ex>
  Main PID: 6366 (nagios)
    Tasks: 9 (limit: 61025)
   Memory: 9.2M (peak: 10.8M)
      CPU: 242ms
   CGroup: /system.slice/nagios.service
           └─6366 /usr/sbin/nagios -d /etc/nagios/nagios.cfg
             └─6367 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
               └─6368 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
                 └─6369 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
                   └─6370 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
                     └─6371 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
                       └─6372 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
                         └─6373 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
                           └─6402 /usr/sbin/nagios -d /etc/nagios/nagios.cfg

Mar 05 18:08:38 vbox nagios[6366]: qh: help for the query handler registered
Mar 05 18:08:38 vbox nagios[6366]: wproc: Successfully registered manager as @w>
lines 1-23
```

github link: [saintZeroCy/nagios](https://github.com/saintZeroCy/nagios)

Reflections:

Answer the following:

1. What are the benefits of having an availability monitoring tool?
 - **Availability monitoring tools help detect system failures early, reduce downtime, improve reliability, and protect business revenue. They send automatic alerts when servers or applications go down, allowing administrators to fix issues before they affect users.**

Conclusions:

In this activity, we successfully installed Nagios Core on both Ubuntu and CentOS using Ansible roles. By automating the setup, we ensured consistency and reduced manual errors. This shows that availability monitoring is important because it helps detect downtime early and protects systems from long service interruptions.